SCENE CLASSIFICATION

Project Proposal Presentation

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OUTLINE

- 1. Introduction
- 2. Timeline
- 3. Results expected
- 4. Descriptors chosen
    - 4.a. GABOR
    - 4.b. CENTRIST
    - 4.c. Color Histogram
- 5. Classifier

INTRODUCTION

Scene classification VS. object recognition

Better results: Multiple feature sets extracted.

1. Separate analysis.
2. Features combined.

TIMELINE

Date	Updates	Details
01/15	**FIRST Update Report**	Start of 3 descriptors - Gabor: selection of optimal parameters, testing - Centrist: calculation of CT values - Color Histogram descriptor. Image dataset selection.
01/22	**SECOND Update Report**	First results of 3 descriptors without particular improvements. - Gabor: formation of the Gabor filter bank, - Centrist: calculation of centrist values - Color Histogram descriptor
01/29	**THIRD Update Report**	First results of 3 descriptors with improvements: - Gabor: feature vector formation, - Centrist - Color Histogram descriptor. Neural Network classifier implementation. Result examination of three approaches individually.
02/05	**FOURTH Update Report**	Hybrid - combined feature vector of three descriptors. Result examination
02/12	Final Project Report	Comparison of approaches. Conclusion.

EXPECTED RESULTS

Achieve:

Create a Scene and Category development with each descriptor:

- Gabor, Centrist, and Color Histogram .

Improve:

The Scene and Category development with a hybrid method

- combining all descriptors for a better image categorization.

Obtain:

A higher accuracy in image categorization with a hybrid method.

- Better than any of the descriptors accuracy used to combine.

GABOR DESCRIPTOR

- Gabor filters possess localization properties in both spatial and frequency domain.
- Frequency and orientation representations of Gabor filters are similar to those of the human visual system
- In the spatial domain, a 2D Gabor filter is a Gaussian kernel function modulated by a sinusoidal plane wave.
- Texture as a feature (texture can be considered to be repeating patterns of local variation of pixel intensities.)
- Gabor filter = texture descriptor

Gabor Descriptor

Feature extraction using Gabor filters consist of the following steps:

1) *configuring Gabor filters in the spatio-frequency domain* - choosing optimal filter parameters (orientation, phase offset, aspect ration, number of directions)

2) *combining Gabor filter bank* that will be applied on images to form feature vectors

CENTRIST VISUAL DESCRIPTOR

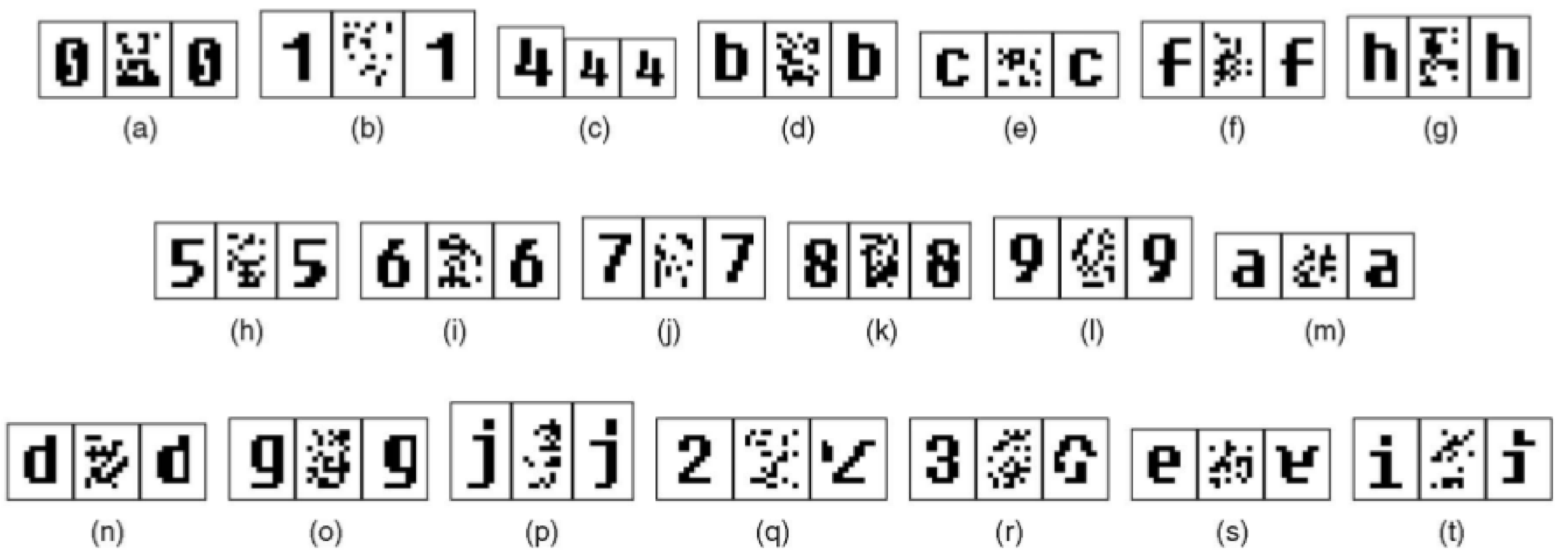
- Calculate CT values by comparing neighborhood,
- Generate histogram = CENTRIST

$$\begin{array}{ccc} 32 & 64 & 96 \\ 32 & \mathbf{64} & 96 \\ 32 & 32 & 96 \end{array} \Rightarrow \begin{array}{ccc} 1 & 1 & 0 \\ 1 & & 0 \\ 1 & 1 & 0 \end{array} \Rightarrow (11010110)_2 \Rightarrow \mathrm{CT} = 214$$

CENTRIST VISUAL DESCRIPTOR

Very good results!

Structure is encoded in the CENTRIST descriptor



COLOR HISTOGRAMS DESCRIPTOR

Bin-ratio provides important information that is collected from ratios between bin values of histograms that are used for scene and category classifications.

<u>These advantages are as follows:</u>

-BRD is robust to cluttering, partial occlusion and histogram normalization.

-BRD captures rich co-occurrence information, and has a linear computational complexity.

-BRD easy to combined with other methods to gather complementary information.

COLOR HISTOGRAMS DESCRIPTOR

Histogram-based representation will capture rich information to be used within the bag-of-words models, where each image will be represented by its histogram of quantized visual words and combined with classifiers for categorization purpose.

Histogram Development and Approaches

- Partial matching information:
Avoid large amount of clutter that brings noises to the object-of-interest.

- Co-occurrence information:
captures correlation information between pairs of histogram bins of visual words

Three recent matching algorithms with high-order statistics:
- Second-order programming,
- Matching
- High order matching.

CLASSIFIER

- As a classifier *Artificial Neural Network* was selected.
- *Tool:* Matlab Neural Network Toolbox
- Neural Network using *feed forward back propagation* algorithm
- *Number of layer*s: to be determined (probably 1 hidden layer)

CONCLUSION

Gabor, Centrist, and Color Histogram descriptors can be easily combined into one method. This combination will give place to a hybrid methodology to gather complementary information for a better categorization and scene classification tasks using a bag-of-words framework.